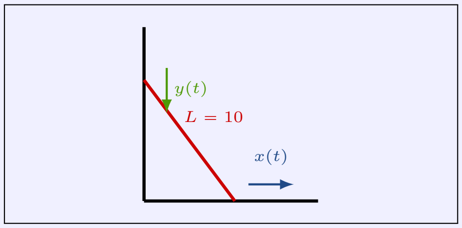
L1.7b Related Rates

§2.8 — what you need to know

The related-rates template

Every problem gets all six boxes, in this arrangement. Worked here on the ladder: *a 10 ft ladder slides down a wall, the foot moving out at 1 ft/s. How fast is the top sliding down when the foot is 6 ft from the wall?*

Picture



Define

C — $L = 10$

V — $x(t),\ \ y(t)$

R — $\dot{x} = +1,\ \ \dot{y} = ?$ *foot moves out, so +*

? $\dot{y}$ when $x = 6$

MEQ: $x(t) \longleftrightarrow y(t)$ $\boxed{x^2 + y^2 = 100}$

Solve:

$\dot{x} = 1, \quad x = 6$

$y = \sqrt{100 - 36} = 8$

$\dot{y} = -\frac{6}{8}(1) = -\frac{3}{4}$

$\boxed{\dot{y} = -0.75 \text{ ft/s}}$

REQ: $\dfrac{d}{dt}\left(x^2 + y^2\right) = \dfrac{d}{dt}(100)$

$2x\dot{x} + 2y\dot{y} = 0$

$2y\dot{y} = -2x\dot{x}$

$\boxed{\dot{y} = -\dfrac{x}{y}\,\dot{x}}$

The numbers go in at Solve, and nowhere earlier. Any value that arrives with a *when* describes one instant; the relation holds for **all** t, so it gets differentiated first.

RATES LINKED BY AN EQUATION

Two rates are **related** when the quantities they measure satisfy an equation. Find that equation — the MEQ — and knowing one rate tells you the other.

A dot means *derivative with respect to time*: $\dot{V} = \dfrac{dV}{dt}$, $\dot{r} = \dfrac{dr}{dt}$, $\dot{\theta} = \dfrac{d\theta}{dt}$. **Use it from the first line.**

$$\boxed{\frac{d}{dt}\left(r^3\right) = 3r^2\,\dot{r}}$$

Every letter is a function of t, so every letter picks up its rate. Related rates **is** implicit differentiation with t underneath the d — the same move as L1.7a, where every y picked up a y'.

AUXILIARY EQUATIONS

Look at the MEQ before you differentiate and count the variables whose rate you do not know. If there is more than one, you probably need an **auxiliary equation** first — a second relationship that removes a variable.

Every variable in the MEQ picks up a rate in the REQ, so the count you can see now is the count you would be stuck with afterwards.

The cone: $V = \frac{1}{3}\pi r^2 h$ has *two* unknown rates, $\dot{r}$ and $\dot{h}$. Similar triangles supply the aux eqn.

$$\frac{2}{4} = \frac{r}{h} \Rightarrow r = \frac{1}{2}h \qquad V = \frac{1}{3}\pi\left(\frac{h}{2}\right)^2 h = \frac{\pi}{12}h^3 \qquad \dot{V} = \frac{\pi}{4}h^2\dot{h}$$

One variable, one unknown rate — and now it solves.

An auxiliary equation holds at **every** instant, so it goes in *early*. A *when* holds at **one** instant, so it goes in *late*. That is the whole distinction.

SIGNS AND RADIANS

Declare the sign of every known rate in the R box, before computing. If it is not obvious, sketch the quantity against t and read the slope: water going in means $\dot{V} > 0$; a shrinking radius means $\dot{r} < 0$.

A negative answer is then **information, not an error** — the top of the ladder really does move down.

Radians are forced when linear and angular measures mix. The searchlight's $\tan\theta = \frac{x}{20}$ puts feet and an angle in one equation, so $\dot{\theta}$ is reported in rad/s and never in degrees per second. With no length in the problem at all you are free to choose.

WATCH OUT

✗ ~~substitute the instant, then differentiate~~ → **the relation holds for all t — differentiate first, substitute at Solve**

✗ $\cancel{\frac{d}{dt}\left(r^3\right) = 3r^2}$ → **r is a function of t, so it is $3r^2\dot{r}$**

✗ ~~a negative rate is an arithmetic slip~~ → **it is the answer — the quantity is going down**

✗ ~~$V = \frac{1}{3}\pi r^2 h$ can be differentiated as it stands~~ → **two unknown rates — find the auxiliary equation first**

✗ ~~$\dot{\theta}$ can be reported in degrees per second~~ → **the problem mixes linear and angular measure, so radians are forced**

✗ ~~the picture is the setup~~ → **the picture is one box of six**